

Unit 7, Lesson 3: Constructing Circle Graphs

Benchmark

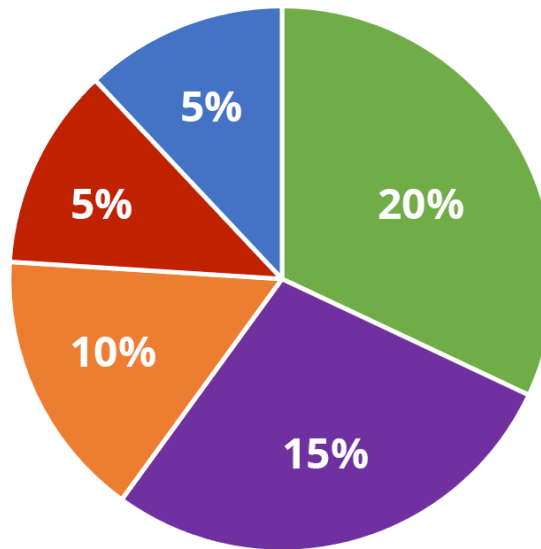
MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

Learning Target

- ☐ I can construct a circle graph given categorical data.

7.3.1 Warm-Up: Data Display

Consider the circle graph.



1. What do you notice about the graph?
2. Explain whether or not this is a valid display of data.

7.3.2 Guided Instruction: Circle Graphs



A **circle graph**, also called a pie chart, is a visual display of categorical data. The whole set of data is represented by the circle and its interior. The categories are represented by fractional parts of the circle.

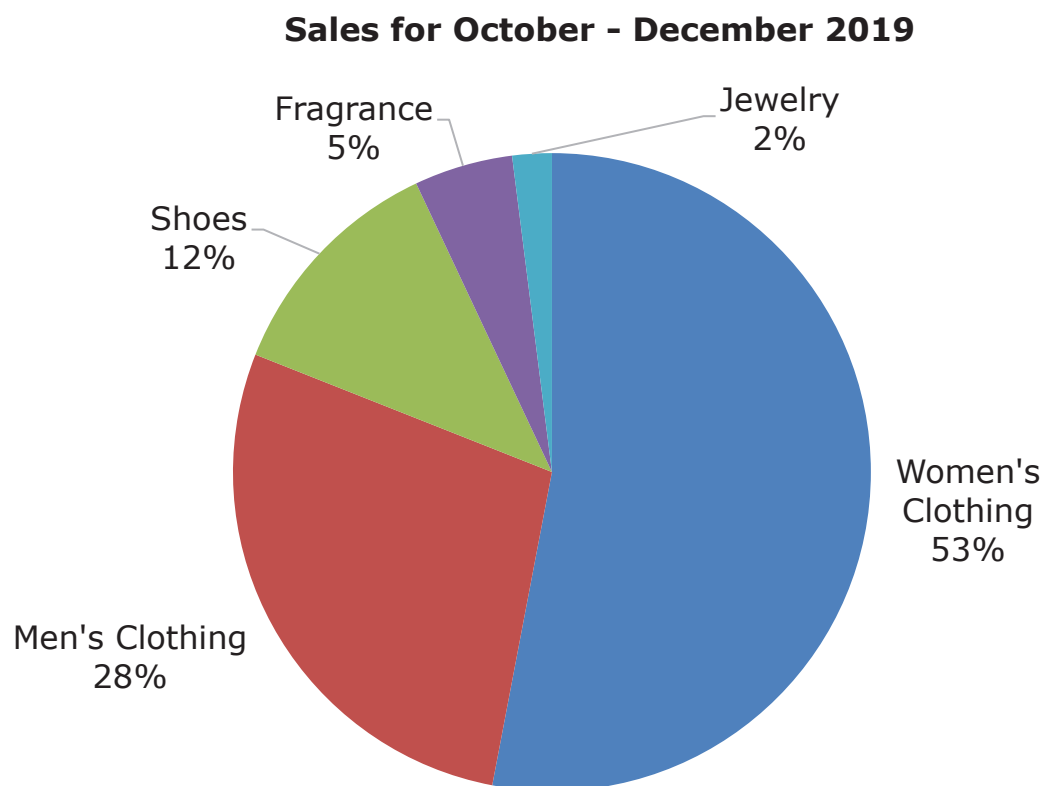
1. The survey data in the table shows data collected in 2019 on the use of different technologies in the United States.

Technology	% of People Who Use It
Internet	86%
Cell Phone	96%
Tablet/Computer	88%

- a. How could someone using more than one of the technologies in the survey affect the percentages?
- b. Discuss with your partner whether the data in the table can be displayed in a circle graph. Summarize your discussion.
2. A store collected data on the percentage of sales that each department generated for the store’s total revenue over three months.

Department	% of Sales Generated
Women’s Clothing	53%
Men’s Clothing	28%
Shoes	12%
Fragrance	5%
Jewelry	2%

- a. With your partner, compare the department store data to the technologies data. Describe any similarities or differences you notice.
 - b. Discuss with your partner why a circle graph is a good display for the department store data.
3. The circle graph displays a store's sales by department over a three-month period.



- a. Describe the information that was included for each sector.

- b. Which department had the largest percentage of sales?
- c. Discuss with your partner how to find the angle measure for each sector.
- d. Find the corresponding angle measure for each department. Round to the nearest whole number.

Department	Angle Measure (°)
Women’s Clothing	
Men’s Clothing	
Shoes	
Fragrance	
Jewelry	

- e. What is the sum of all five calculated angle measures?
- f. Use your protractor to compare your calculated values with the degrees of each sector in the actual circle graph. Then, discuss with your partner how they compare.

7.3.3 Collaboration: Constructing a Circle Graph

Complete the following with your partner.

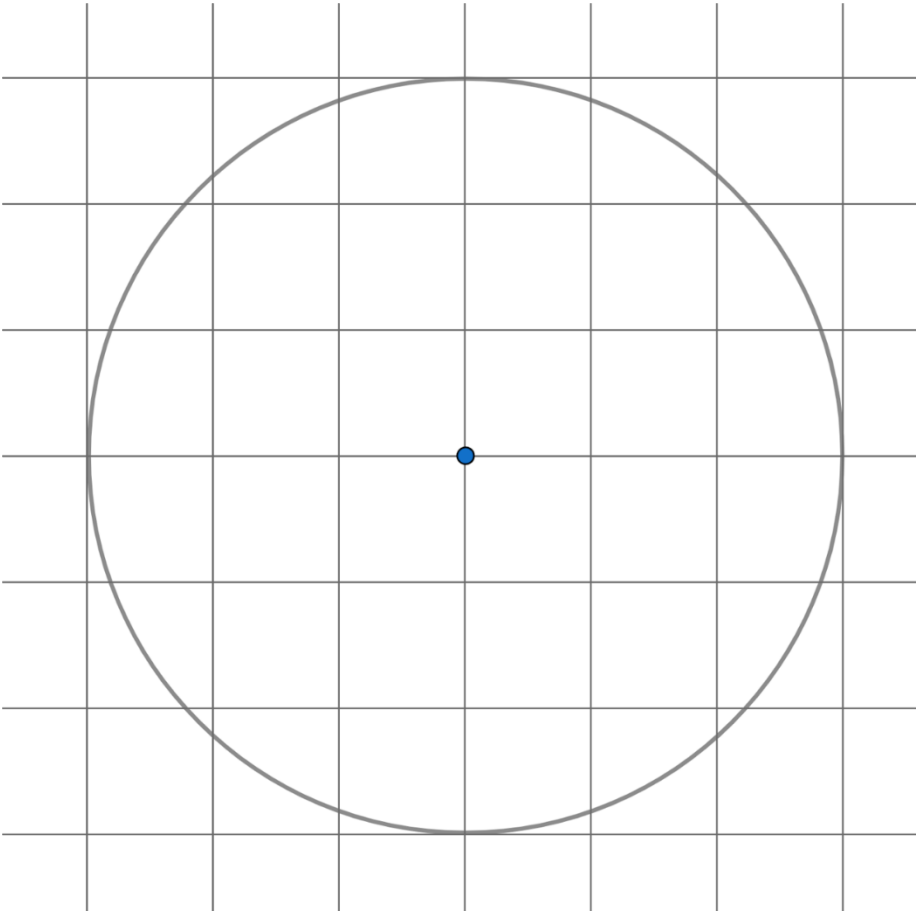
1. A travel agency sent out a survey to find out which local airports people preferred. The travel agency received 380 responses and compiled the data into the table shown.

Airport	Number of People Who Selected the Airport
Northshore	206
Southport	122
East Town	26
West Lane	26

- a. Use the data to complete the table. Round percentage to the nearest whole number and angle measure to the nearest tenth.

Airport	$\frac{\text{airport choice}}{\text{total responses}}$	Percentage	Angle Measure (°)
Northshore	$\frac{206}{380}$		
Southport			
East Town			
West Lane			

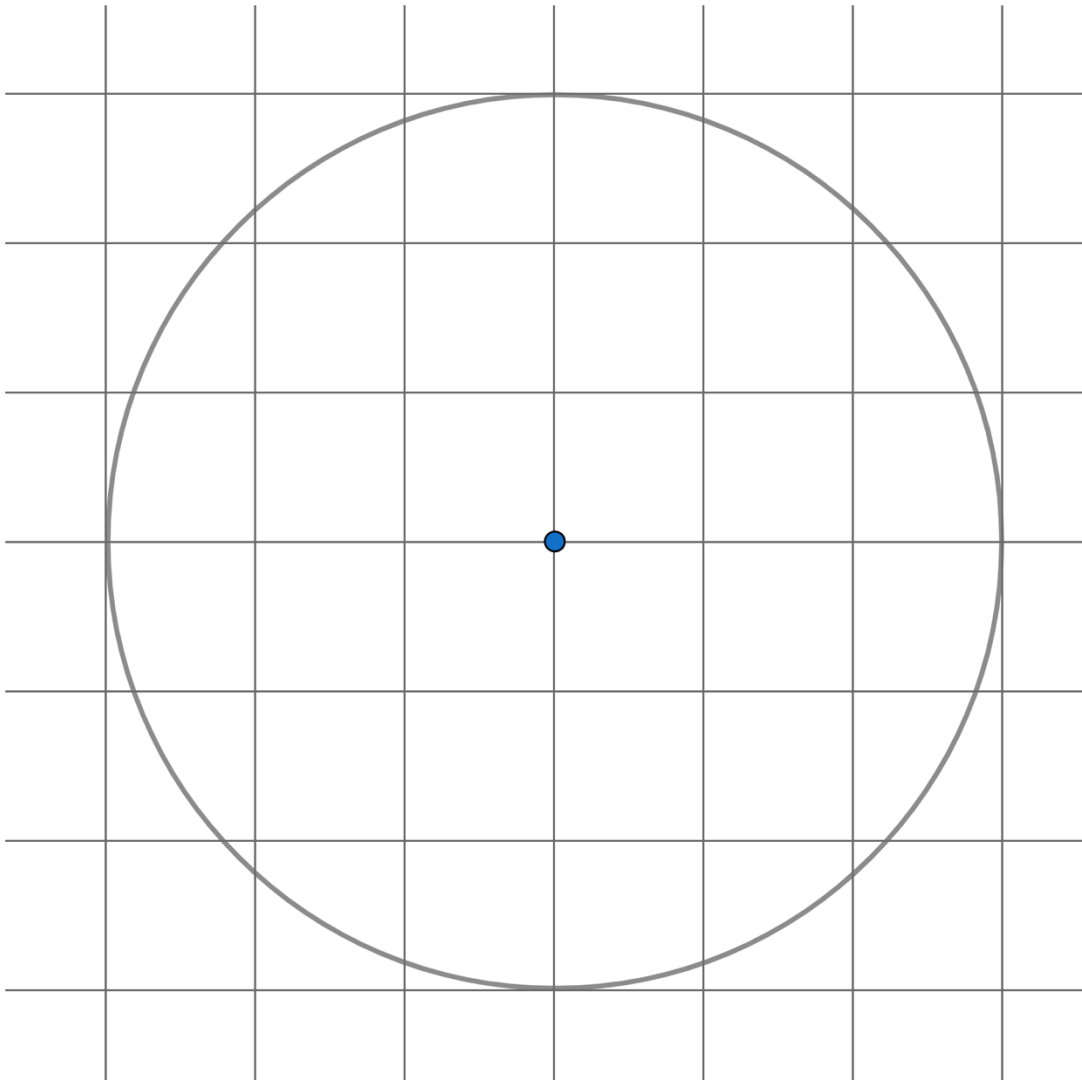
b. Use the data in the table and a protractor to construct a circle graph.



2. A restaurant is developing a new menu, so they decide to survey people’s favorite local seafood. The results of the survey are shown in the table.

Favorite Seafood	Percentage Preferred
Grouper	15%
Red Snapper	39%
Stone Crab	30%
Pink Shrimp	16%

- a. Use the data and a protractor to construct a circle graph.



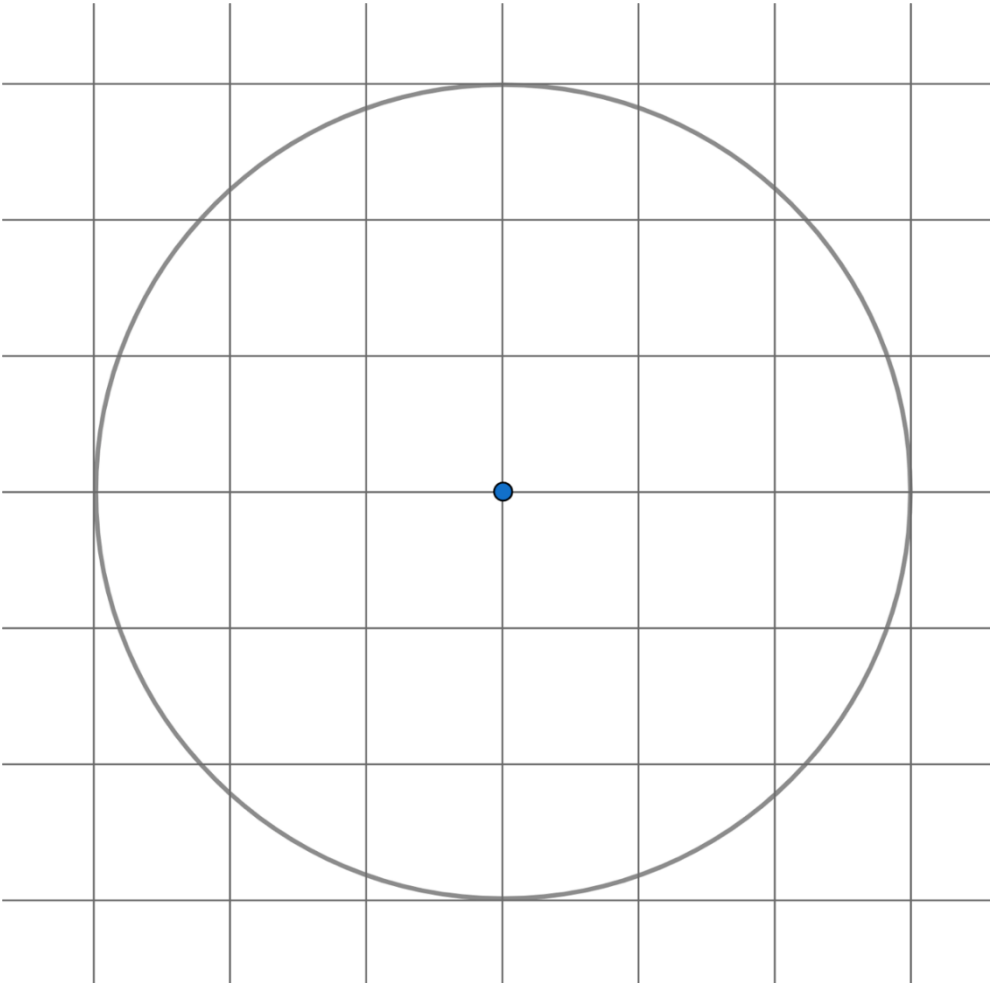
- b. Summarize the process you and your partner used to construct the circle graph.

7.3.4 Your Turn!

1. The Pasco Middle School Pirates are conducting a survey to determine their students’ favorite Florida sports team name. The results of the survey are shown in the table.

Florida Team Name	Percentage Preferred
USF Bulls	19%
Miami Hurricanes	7%
Florida Gators	32%
UCF Knights	22%
Florida State Seminoles	20%

Use the data and a protractor to construct a circle graph.



7.3.5 Wrap-Up: Ticket Out the Door

1. The Rudolph Family’s top five monthly expenses are as follows:
- Rent \$950.00

• Utilities \$236.00

• Groceries \$800.00

• Gas/Car Insurance \$356.50

• Miscellaneous \$500.00

a. Complete the table. Round each percentage to the nearest whole number and angle measure to the nearest tenth.

Expense	$\frac{\text{expense category}}{\text{total expenses}}$	Percentage	Angle Measure (°)
Rent	$\frac{950}{2842.50}$		
Utilities	$\frac{236}{2842.50}$		
Groceries	$\frac{800}{2842.50}$		
Gas/Car Insurance	$\frac{356.50}{2842.50}$		
Miscellaneous	$\frac{500}{2842.50}$		

- b. Construct the corresponding circle graph.

